

Non-Invited Review

Current approaches to avoid the culling of day-old male chicks in the layer industry, with special reference to spectroscopic methods

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ABSTRACT The negative correlation between fattening and laying performance prevents breeding improvement in both laying performance and meat yield. Therefore, specialized chicken lines have been bred in order to achieve either an efficient production of high-quality eggs or high growth rates. As a result, day-old male chicks are culled in the layer hatchery, which poses animal welfare and ethical problems. Breeding companies, scientific groups, and hatcheries are attempting to resolve this issue, with a common aim to find feasible alternatives for the routine killing of male layer chicks. Some approaches aim to influence the sex ratio, while others target at the economically feasible use of the male layer offspring, such as the

fattening of “laying hen brothers” or crossbreedings of layers and broilers to create “dual-purpose chickens.” Another approach is the sex determination prior to hatch. One of the prerequisites of *in ovo* sex determination is a practicable method that can be used in industry. The analysis needs to be rapid, cost-efficient, and highly precise; in addition, negative impacts on hatching rate, animal health, and/or performance parameters should be limited. Furthermore, sex determination should be performed before the sensory nervous system’s response of the chick embryo to certain or potentially harmful stimuli is developed, which according to current knowledge is before the d 7 of incubation.

Key words: day-old male chick, culling, animal welfare, *in ovo* sex determination, sexing

2018 Poultry Science 97:749–757
<http://dx.doi.org/10.3382/ps/pex389>

INTRODUCTION

As a highly negative correlation exists between fattening and laying performance, specialized chicken lines have been bred over time that either efficiently produce high-quality eggs or have high growth rates. While both males and females are fattened in broiler production, there is currently no economically worthwhile use of the male offspring of layers. Therefore, day-old male chicks are culled in the layer hatchery, which poses both animal welfare and ethical issues (Ort, 2010; Buhl, 2013). For some time now, there have been ongoing investigations aiming at the development of animal welfare-compliant strategies that make the killing of day-old male layer chicks either avoidable or at least performed

in an ethically acceptable and economically feasible manner. During the workshop “Management of newly hatched male chicks from layers” in 2002, participants discussed the current knowledge as well as potential solutions (Damme and Ristic, 2003; Ellendorff and Klein, 2003; Gerken et al., 2003; Hardy, 2003; Kagami, 2003; Nandi et al., 2003; Phelps et al., 2003; Preisinger, 2003; Seemann, 2003; Tiersch, 2003; Klein et al., 2003a,b). However, so far none of the potential solutions has advanced to the level of practical application. In the current review, different approaches for avoiding the culling of day-old male chicks are discussed.

SEX DETERMINATION IN BIRDS

In birds, contrary to humans and mammals, males are homogametic with 2 Z sex chromosomes, whereas females are heterogametic with 1 Z and 1 W sex chromosome. Sex chromosomes can be heteromorphic as in carinate birds or homomorphic as in the ratites (Smith

© 2017 Poultry Science Association Inc.
Received August 25, 2017.
Accepted November 17, 2017.

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et al., 2007). All autosomes in birds are identical for males and females. While the gender-determining chromosome in birds is located within the egg, sex ratio in birds is not manipulable by sperm sexing and subsequent sorting. Göth and Booth (2005) described a temperature-dependent sex ratio in birds in the Australian brush-turkey (*Alectura lathamii*). Molecular sexing of chicks and embryos of Australian brush-turkeys confirmed that male embryo mortality was greater at high temperatures while female embryo mortality is greater at low temperatures in this species (Eiby et al., 2008). To date, there is no evidence of similar effects on sex ratio or of a temperature-dependent sex determination in the chicken (Collins et al., 2013).

APPROACHES TO INFLUENCE SEX RATIO

Aslam et al. (2013) sought to correlate the sex of the egg in unincubated eggs to a wide array of egg components (i.e., yolk concentrations of testosterone, estradiol, androstenedione, progesterone, dihydrotestosterone, and glucose, as well as egg weight and dimensions) and the hens' body weight. In addition, they also studied the relationships among all measured parameters. Associations were established between some yolk hormones (progesterone associated with testosterone, estradiol, and androstenedione; androstenedione with testosterone; dihydrotestosterone with estradiol and androstenedione) as well as between yolk testosterone and egg length or egg weight, respectively. No significant overall differences between male and female chicken eggs in any of the measured egg parameters were found. However, some correlations were observed, such as between the sex of the egg and dihydrotestosterone and with hen body weight, which predicted estradiol levels, and also between estradiol levels and egg width for predicting the gender of the egg.

In a different study, Aslam et al. (2014) substantially elevated blood plasma corticosterone levels through corticosterone feeding and studied the primary offspring sex ratio (defined as the proportion of male fertile eggs determined in freshly laid eggs, i.e., without egg incubation). Mean plasma corticosterone concentrations were appreciably higher in the treatment group but were not associated with fertility rate, sex ratio, and laying rate. This treatment by itself affected sex ratio as well as laying rate and fertility rate in interaction with hen body mass, but did not affect egg sex. While body mass was negatively associated with laying rate, sex ratio, and fertility rate per hen in the corticosterone group, it had a positive association with sex ratio in untreated hens. These interactions were already evident when taking the body mass at the very start of the experiment, indicating the existence of intrinsic differences between light and heavy hens as far as their reaction to corticosterone treatment goes. The effects on laying rate, fertility rate, and sex ratio suggest that some factors

related to body mass act together with corticosterone to modulate ovarian functions.

In yet another study, Aslam et al. (2015) induced a decrease in body condition and egg mass by implementing feed restriction in laying chickens. This led to an overall decline of egg mass. With more severe feed restriction and a steeper decline of egg mass in the second period of treatment (d 9 to 18), the sex ratio per hen (proportion of male eggs) showed a significantly negative association with mean egg mass per hen. Based on this association, 2 groups of hens were selected from the feed restriction group, that is, hens producing male bias with low egg mass and hens producing female bias with high egg mass with overall sex ratios of 0.71 and 0.44, respectively. For these 2 groups of chickens, Aslam et al. (2015) found no significant differences regarding the expressed genes. However, it was shown using gene set enrichment analysis that a number of cellular processes related to cell cycle progression, mitotic/meiotic apparatus, and chromosomal movement were enriched in female-biased hens or high mean egg mass as compared with male-biased hens or low mean egg mass.

USE OF MALE LAYERS FOR MEAT PRODUCTION

Fattening of "Laying Hen Brothers"

To date, there have been different strategies described to avoid culling the male chicks. Koenig et al. (2010, 2012a,b) performed studies on commercial broilers (Ross 308) and different genotypes of laying-type cockerels: medium heavy, brown-eggshell Lohmann Brown (**LB**) and Hy-Line Brown (**Hyline**); light, white-eggshell Lohmann Selected Leghorn (**LSL**) and Dekalb White (**Dekalb**). The cockerels were fed standard diets ad libitum and were reared on deep litter. The broilers attained the intended carcass weight of about 650 g after 19 d, the laying-type cockerels after 47 d (LB, Hy-Line) or 49 d (LSL, Dekalb). The results on growth performance showed that it was reasonable to stop fattening duration at this point. Feed conversion was calculated to be 1:1.2 and 1:2.45 for broilers and egg-laying types, respectively. The weights of valuable parts (i.e., breast, legs) were higher for the former than for the latter. It must therefore be concluded that up to now the fattening of "laying hen brothers" does not pay off, as they require more time and food to grow and the pectoral muscle, which is preferred by the consumer, is not comparable to that of a broiler chicken. Furthermore, the carcasses do not match the consumer's vision of an ideal "plump fryer," and require special ways of preparation and cooking. In addition, marketing of "spring chicken" or "laying hen brothers" is challenging, so the fattening of male layer chicks and their marketing must still be considered as a niche production (Koenig et al., 2010, 2012a,b).

Dual Purpose Chicken

Another strategy is the crossbreeding of layers and broilers to create “dual-purpose chickens” (e.g. Lohmann Dual) as a commercially marketed breeding line, thus gaining a compromise between meat and egg production. Lohmann Dual birds consume up to 30 g more per d, and therefore feed costs are calculated to be up to 50% higher for the entire laying period than for commercial layers. Additionally, Lohmann Dual hens not only laid a fewer number of eggs, but also showed a smaller egg size, thereby lowering egg mass output in addition. The eggs laid by Lohmann Dual chickens have a light-brown eggshell color. It was argued that these economic disadvantages and ecological imbalance of using more feed to produce less high-quality protein-food have to be accepted to take advantage of the higher meat production of dual-purpose birds compared to layer males. Compared to a slow-growing broiler, the live weight gain in Lohmann Dual is moderate. From wk 3 until 10 wk of age, dual birds and broilers grow further apart. In the fattening period, at 8 wk of age, male dual-purpose birds have a live weight of just about 2 kg, whereas a slow-growing broiler counterpart typically has a bodyweight of 3.2 kg.

Fed with broiler diets for 70 d, the dual cockerels reach a live weight of 3 kg, and a carcass weight of about 2 kg. In terms of carcass performance, there are hardly any differences between the dual cockerels and conventional broilers. The amount of valuable parts was at 50%. Unlike special broiler lines, the dual cockerels have a much lesser portion of breast meat in favor of the portion of the thighs. However, to date there have been no viable solutions to avoid significant economic losses, particularly regarding their food utilization and efficiency, when compared to specialized layers or broilers (Icken et al., 2013; Preisinger et al., 2014).

IN OVO SEX DETERMINATION—NON-OPTICAL METHODS

Due to ecological and economic reasons, *in ovo* sex determination of specialized high-performance layer lines is currently the preferred method to avoid the culling of day-old male layer chicks. Over the past few years, several different approaches have been pursued in order to establish a method suitable for practical use in the hatchery.

There are several prerequisites for achievement of a sex determination method in chickens feasible on an industrial basis (Kaleta and Redmann, 2008). The analysis needs to be rapid, cost-efficient, and highly precise, and must not have any considerable negative impacts on hatching rate, animal health, and/or performance. Additionally, sex determination needs to occur before pain perception has evolved in the chick embryos (Krautwald-Junghanns et al., 2014). The ability of bird embryos to experience *in ovo* nociception is no longer in question (Bjørnstad et al., 2015). The first sensory

afferent nerves develop in the chicken embryo on the d 4 of incubation, but a synaptic connection to the spinal cord is not present before d 7 of incubation, which makes nociception impossible in the first third of incubation (Eide and Glover, 1995, 1997). Therefore, no sensitivity of the chick embryo is to be expected before d 7 of incubation (Rosenbruch, 1994, 1997; Aleksandrowicz and Herr, 2015).

Morphometric Studies on Outer Shape of the Eggshell

Imholt (2010) measured the maximum length and the maximum diameter of a total of 1,223 eggs of 6 commercial layer breeds and 6 fancy breeds for the comparison of genetically female and genetically male birds. Investigating a possible connection between the outer shape of the eggshell and the sex of the chick therefrom, the arithmetic mean of maximum length and maximum diameter were calculated for every single egg and subsequently compared with each other. While measurements showed that the outer shape of eggshells differed, it was not possible to correlate those differences to the sex of the developing chick with the used methods.

Yilmaz-Dikmen and Dikmen (2013) carried out a similar study to determine the sex of fertilized white layer eggs by using morphological measurements. Before incubation, egg length, width and weight were measured of a total of 300 white layer eggs. Eggs were incubated and sexed at the end of incubation period. The egg volume and shape index were estimated by using these measurements for each egg. The effect of egg weight and replicate number was not significant on the sex of the hatching chick. Depending on the hatching chick's sex, the effects of egg shape index, egg length, egg width, and volume of the egg differed significantly. According to the results of this study, the authors assumed that morphological measurements of the pre-incubated egg might be an indicator of the hatching chick's gender.

Egg Odor

Webster et al. (2015) reported that volatiles from developing eggs of Japanese quail (*Coturnix japonica*) carry information on egg fertility, along with the sex and developmental status of the embryo. Specifically, it was shown that egg volatiles undergo change over the course of incubation, and differ not only between fertile and infertile eggs, but were also sex-predictive as early as d 1 of incubation. One of the volatile ketones that differed between eggs containing male and female embryos (2-undecanone) has previously been identified as a hormone-linked constituent of avian odor (Whittaker et al., 2011). To date, however, there is no corresponding data on the likelihood of an *in ovo* sex determination in chicken based on odor attributes.

Molecular Sexing Assays

Using a small amount of crude material, Clinton et al. (2016) developed a simple and robust procedure that permits rapid identification of the sex of individual embryos. The sexing assay is based on Hologic Invader[®] technology, an isothermal “PCR-free” approach which takes advantage of a thermostable structure-specific archaeobacterial flap endonuclease (**FEN**) that cleaves nucleic acid molecules at specific sites, based on structure rather than sequence. Applying this procedure, sex can be determined in 5 to 15 min using either tissue fragments, small volumes of whole blood, or a small number of isolated cells. In all instances, the Hologic Invader[®] assay results agreed with the sex of embryos as determined by an established protocol. However, up to now the method was only developed for use under laboratory conditions.

Genetic Engineering

The production of transgenic chickens has increasing applications in biotechnology providing excellent model organisms for developmental biology research and bioreactors for pharmaceutical proteins. Very likely the application with the greatest global impact will be enhancing the security of chicken meat and egg production by generating chickens resistant to disease and/or featuring improved production traits (Tyack et al., 2013). The genetic marking of sex chromosomes is also discussed as a possible route for *in ovo* sex determination in chicken. Studies by Quansah et al. (2013) and Doran et al. (2017) focused on the production of genetically engineered hens, and described the marking of the Z chromosome of breeding hens with green fluorescent protein. This method was successfully used for sex determination in layers, with the gender being deducted from sex-specific patterns of germinal disc fluorescence in non-incubated eggs (Bruijns et al., 2015).

Evaluation of Hormon Concentration in the Allantoic Fluid

Evaluating the concentration of hormones in the allantoic fluid for sexing provides reliable results in later developmental stages of embryonic development only (Phelps et al., 2003; Tran et al., 2010), at a time at which the embryo is likely to already feel pain. Weissmann et al. (2013) established a method for *in ovo* sex identification on d 9 of incubation by measuring estrone sulfate in the allantoic fluid. It was observed that male embryos displayed significantly lower hormone levels in the allantoic fluid compared to females. Predictive sexing accuracy was above 98% for *in ovo* sexing on d 9. Compared to an untreated control group, the hatching rate of the experimental group was reduced by 1.4 to 3.5 points of percentage (brown layers) and 12.7 points of percentage (white layers) due to sam-

pling of allantoic fluid. For both groups, the hatching weight of the day-old chicks was the same. Further monitoring of the post hatching performance revealed that the use of allantoic fluid has negligible impact on the hens. Although distinctions in weight of control and experimental groups were observed during the rearing period, the adult hens' laying performance, egg and body weight did not differ significantly between the groups.

IN OVO SEX DETERMINATION—OPTICAL AND IMAGING METHODS

Other than morphometric, molecular biological, or biochemical methods of sex determination, there have been different optical and imaging methods successfully performed in birds. A great advantage of optical methods is their contactless application.

Reflectance Spectroscopy and Hyperspectral Imaging

Reflectance spectroscopy in combination with statistical data analysis was performed on 450 White Leghorn eggs from a young flock (24 wk of age). The eggs were measured on d 0, 1, 2, and 10. The data set underwent PCA discrimination using the Unscrambler platform and a neural-network classification model formulated for fertility and gender. Actual fertility was measured on d 10 of incubation, and the final sorting of gender was conducted at hatch. A comparison of actual and predicted results indicated that prediction capability is over 95% for fertility tested on d 0 and 90% for gender detection on d 10. Rozenboim and Ben Dor (2011) thus judged the reflectance spectroscopy method to be adequate for detection of gender of chicken embryos at mid-incubation period.

Leiqing et al. (2016) described the use of hyperspectral imaging for *in ovo* sex determination, which uses a sensing system capable of plotting high numbers of closely spaced wavelengths. However, by using different mathematical techniques, there was only a 75.0% to 82.9% chance of predicting the correct sex, and this method could only be used on d 10 of incubation. Göhler et al. (2017) described a non-destructive optical technique for sex determination in layer lines with sex-specific down feather color. The accuracy of sex determination was evaluated for 11- to 14-day-old embryos. Applying this method, the sex of the chicken embryo can be determined on d 14 of incubation with an overall accuracy of approximately 97%.

Fourier Transform Infrared Spectroscopy

Sex determination using Fourier transform infrared (**FTIR**) spectroscopy is already feasible in the non-incubated egg. This is because the germinal disc of a freshly laid and fertilized chicken egg is composed of between 40,000 to 60,000 blastoderm cells, which contain

genetic information and thus can be used to determine gender (Steiner et al., 2011). Several imaging methods have been employed to locate the germinal disc in previous studies (Klein et al., 2002; Bartels et al., 2008; Burkhardt et al., 2011). However, the eggshell poses an impenetrable barrier for optical analysis; therefore, optical access is critical for spectroscopic gender determination. An opening of the eggshell is achieved with a suitable CO₂-laser at the pointed egg pole, thus providing a precisely circumscribed ablation in the calcified shell in a fraction of a second. The circular movement of the highly focused laser beam creates a predetermined breaking point, leaving just a thin connecting bar in the calcified shell. This prevents the high-energy laser radiation from entering the egg and harming the early embryo. Currently, a shell fenestration of 12 mm in diameter is required for this process. Effects of egg windowing at different time points on hatching rates were observed in a breeding experiment, where a total number of 4,736 eggs was divided into several groups (one control group [n = 2,211] and groups in which an egg shell opening was achieved after 0 h [n = 515], 24 h [n = 801], and 72 h [n = 1,209] of incubation, respectively). The most notable result was that in non-incubated eggs, opening of the shell resulted in a drastic reduction of the hatching rate (6.6%). In contrast, when eggs were incubated for 72 h, the same manipulations had considerably less effects on embryonic development with a hatching rate of 80.9% (Bartels et al., 2014). The blastoderm appears to be highly sensitive to any environmental changes during that time, as it has already been described in the literature (Fineman et al., 1986).

Raman Spectroscopy

Raman spectroscopy, another type of vibrational spectroscopy, uses monochromatic light to illuminate the object under examination. The spectrum of scattered light is analyzed following its interaction with the sample. Raman spectra are unique for each molecule and are often referred to as a “molecular fingerprints.” As the biochemical composition of cells of female and male birds is slightly but significantly different, Raman spectroscopy allows *in ovo* sex identification based on the spectral signature of germinal or blood cells (Harz et al., 2008; Galli et al., 2016).

By choice of a near-infrared (NIR) excitation wavelength (e.g. 785 nm), damaging of live cells can be avoided since photons in NIR do not carry enough energy to induce molecular changes. NIR Raman spectroscopic studies were used to find differences in nucleated blood cells in a study conducted by Galli et al. (2016). Eggs were incubated for 80 to 88 h until an extraembryonic vascular system was established. Hence, eggs were opened at the pointed end using the CO₂-laser method described above. It was found that the air pocket located at the blunt end of the egg gradually decreases after opening of the shell. As a result,

the embryo changes position, resulting in a downward movement of about 5 mm occurring within the first few minutes after opening. Regardless of this, the vascularized area of the embryo keeps floating above the yolk and remains on the surface. The hatching rate for eggs with a shell window of 10 mm in diameter was determined at 95% in comparison to eggs without perforation. A 2-mm larger perforation led to a slightly lower hatching rate of ~ 91%. However, the diameter of the shell window required for measurement is primarily determined by the optical accessibility with a high numerical aperture objective. In the studies described by Galli et al. (2016), a shell window of 12 mm in diameter consistently enabled vessel sampling without clipping the laser beam. In order to display the embryonic vessels, a camera system was developed which automatically selects a suitable vessel within the aperture and positions the focus of the laser beam. During spectroscopic measurement, the vessel is held in place by automated tracking (auto-focusing/tracking).

The excitation of Raman scattering is performed by a diode laser emitting a wavelength of 785 nm, which is connected to the microscope by a 100 μ m optical fiber. The laser is then focused on the samples using a long working distance microscope objective. The precise focus of the excitation laser on a blood vessel is essential for high accuracy of the spectroscopic *in ovo* sexing. Although the recorded *in ovo* spectra appear to be distinguishable in terms of the early embryo's gender, a relatively broad, overlapping range will lead to a low accuracy of spectral-based classification. As variation of the spectral signals is quite large, it is not possible to classify the spectra by a simple evaluation of band intensities. More sophisticated methods for data analysis and supervised classification have to be applied. The obtained spectra were analyzed by chemometric methods as a superposition of high fluorescence intensity and weak Raman bands, which correspond mainly to hemoglobin, lipids, and nucleic acids. The classification process relies on the fact that the sex chromosomes of carinate bird species are different. This can also be found in the chicken, where the total amount of DNA is about 2% higher in males (Steiner et al., 2011).

The recorded Raman signal is plagued by a strong background fluorescence signal, mainly originating from hemoglobin (Chaiken et al., 2009). Further observations showed that spectral analysis of the near-infrared fluorescence signal of blood flowing within extraembryonic vessels can indeed provide information on the sex of domestic chicken eggs (Galli et al., 2017a,b). Figure 1 shows typical *in-ovo* Raman spectra obtained from male and female embryos. Clear spectral differences are marked and assigned to molecular groups. Spectra of male embryos exhibit stronger phosphodiester linkage stretching vibrations of nucleic acids and C–C stretching modes, respectively. Spectra of female embryos show slightly stronger amide III and CH_x deformation modes. The median fluorescence signal was stronger for male embryos compared to females, which may be due to a

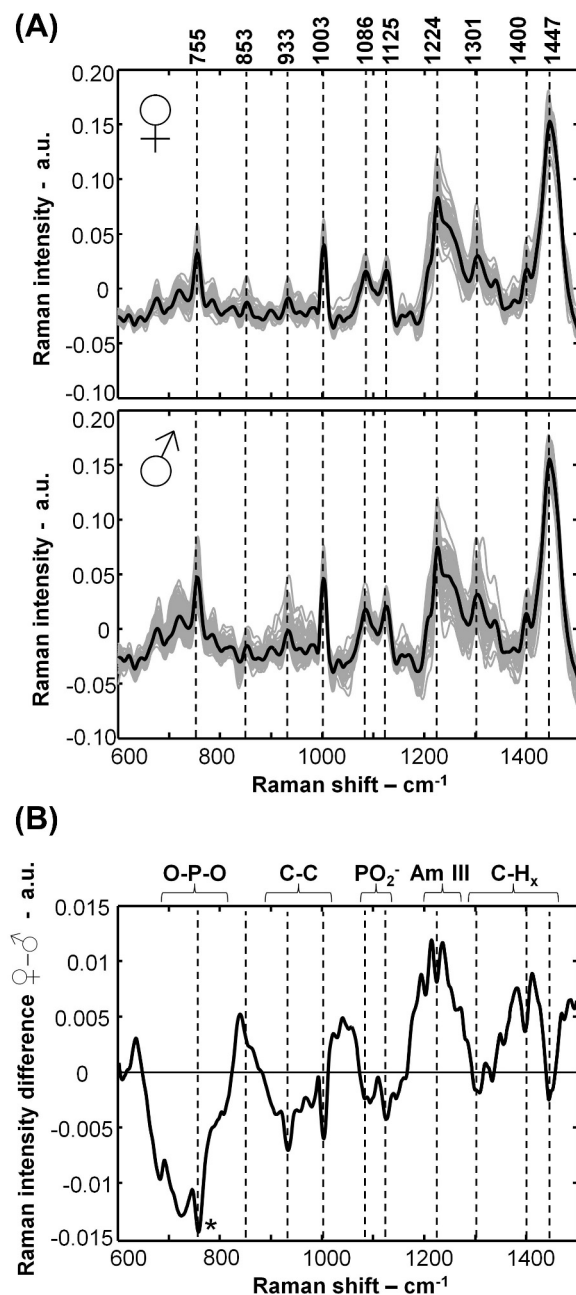


Figure 1. Raman spectra of embryonic chicken blood at d 3.5 of incubation. A: Raman spectra after preprocessing and elimination of the fluorescence signal. B: Difference Raman spectrum obtained by subtracting the mean male spectrum from the mean female one. (Reproduced with permission from Ref [Galli et al., 2016]).

higher hematocrit proven at d 13, 15, and 18 of incubation (Morita et al., 2009).

As both NIR-excited Raman analysis and fluorescence spectroscopy are harmless and contact-free, the risk of contamination is very low. The *in ovo* fluorescence signals of embryonic female and male samples are represented in Figure 2. Although spectra of male embryos on average show higher fluorescence intensities than those of female ones, highly sophisticated methods of data analysis have to be performed to achieve a reliable spectral classification. Raman and fluorescence spectroscopy was also used to determine the sex

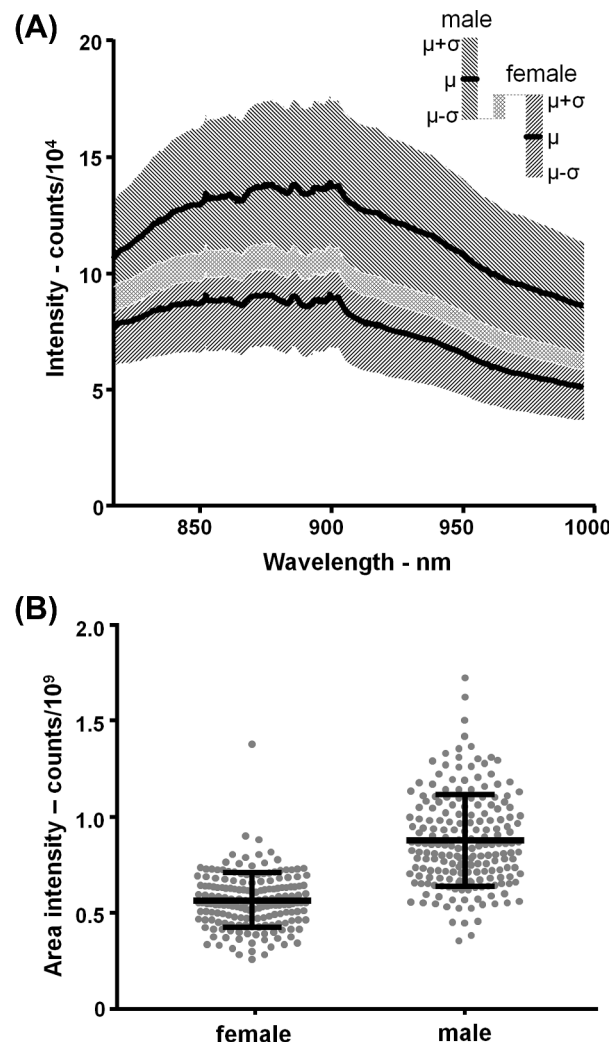


Figure 2. A: In-ovo spectra of embryonic blood acquired from female and male eggs. The overlapping range of standard deviations is highlighted in light gray. B: Total area intensities calculated from spectra shown in Figure A. The horizontal black lines indicate mean values and standard deviations. (Reproduced with permission from Ref [Galli et al., 2017a]).

without removing the inner egg shell membrane. Egg sexing based on supervised classification of the spectral features attained an overall correct rate of 91% (Galli et al., 2017c). This simplifies the whole process automatization and offers the best premises for deployment in the layer industry.

The openings in the female eggs are sealed using biocompatible adhesive tape (3 M Durapore™, 3 M Deutschland GmbH, Germany), and these eggs can then be further incubated until hatch.

Magnetic Resonance Imaging

Davenel et al. (2015) tried to develop a non-invasive method based on magnetic resonance imaging (MRI) for an *in ovo* sex determination in domestic chicken. Besides the embryo, the MRI sequence, particularly the T1 weighed images, allowed to differentiate very clearly albumen, vitelline sac and the allantoic and amniotic

cavities, but their measurements did not highlight significant differences between both sexes.

DISCUSSION AND CONCLUSIONS

Worldwide, researchers have sought for a solution to the animal welfare problem of the culling of approximately 7 billion day-old male layer chicks per year (Poultry Site, 2015), with this issue not only being present in the scope of conventional egg production (cage, barn, and free range eggs), but also in the ecological production of laying hens. Although different approaches have been used in the past to replace culling by sexing the chicks before hatching, no method has proven to be suitable for everyday use (Aerts et al., 2009).

Some of the currently proposed approaches to prevent the culling of male chicks are not practicable for large-scale application, either due to economic aspects, low acceptance by the consumer (e.g., dual-purpose chickens or the production of transgenic animals) or insufficient precision. For example, approaches to influence hens to produce more female eggs showed that while sex ratio can be influenced, the practical application is uncertain and currently impracticable (Aslam et al., 2013, 2014, 2015). The measurement of hormone levels in the allantois can only provide results after 9 to 10 d of incubation at the earliest (Phelps et al., 2003; Tran et al., 2010; Weissmann et al., 2013, 2014), and at this time the embryo most likely has already developed the ability to feel pain. Despite this, neither the problem of culling nor the marketing of male embryos at this late developmental stage has yet been brought to a practical solution. Culling by use of CO₂ through the egg shell would be prolonged and unreliable (Krautwald-Junghanns et al., 2015).

The optic method presented is conducted in a much earlier phase of incubation. Considering that an extraembryonic blood vessel system has already developed within 3 d of incubation, the genetic information contained in blood cells (which contain a nucleus in birds) can be used for contactless sex determination (Galli et al., 2016, 2017a,b,c). By focusing a stimulating laser on a small vessel, achievement of sex determination is currently possible with a specificity/sensitivity of >95%.

The advantages of the presented spectroscopy and/or fluorescence approach not only lie in its high precision and short analysis time, but also in its opportunity to utilize early embryonic stages for analysis. Especially the exploitation of fluorescence bears the potential to develop industrial systems for egg sexing which are not based on expensive spectrometers, but just make use of few light detectors with suited bandpass filters to measure the signal intensity in selected spectral ranges. Eggs determined as male can immediately be removed from the egg trays after sexing. This would, first, result in an improved utilization of loading capacity in the incubators at an early stage, and second, there would

be a proportionately greater yield of male eggs containing high-quality proteins and relatively small amounts of embryonic tissue after 80 to 88 h of incubation. Depending on national regulations, these could be sold for use as quality end-products (e.g., fish feed).

However, in a practical view, the use of Raman and fluorescence spectroscopy depends on additional studies considering parameters as egg size, egg age, and egg storage conditions, as these factors could possibly affect the accuracy of spectroscopic *in ovo* sex determination.

A disadvantage of most methods for sex determination in eggs is the required perforation of the egg shell. Up to now, NIR Raman spectroscopy as well as fluorescence spectroscopy necessitates a 12 mm diameter opening in the egg shell, which can be generated by a CO₂ laser. Overall, the effect of the laser used for shell perforation is negligible. However, the manual levering of the shell results in a reduction of hatching rate of about 10%. Manual opening can impair embryonic structures and lead to developmental disorders or embryonic death. Furthermore, it cannot be applied under practical conditions in the future. Further research on the development of methods for a fully automated shell opening process is currently ongoing.

In ovo sex determination has a true potential to end the culling of male chicks in the egg-producing industry, but full automatization of the processes to guarantee high sexing speed and fulfill industrial demands is required to allow transfer of this technology inside the hatcheries in the near future (Galli et al., 2017b).

ACKNOWLEDGMENTS

The project was supported by funding from the Federal Ministry of Food and Agriculture (BMEL), based on a decision by the Parliament of the Federal Republic of Germany via the Federal Office for Agriculture and Food (BLE), under an innovation support program (grant no. 2813IP003).

REFERENCES

- Aerts, S., R. Boonen, V. Bruggeman, J. De Tavernier, and E. Decuypere. 2009. Culling of day-old chickens: opening the debates of Moria? Pages 117–122 in *Ethical Futures: Bioscience and Food Horizons*, K. Millar, P. Hobson West, and B. Nerlich, eds. Wageningen Academic Publishers, Wageningen, the Netherlands.
- Aleksandrowicz, E., and I. Herr. 2015. Ethical euthanasia and short-term anesthesia of the chick embryo. *Altex* 32:143–147.
- Aslam, M. A., M. Hulst, R. A. H. Hoving-Bolink, M. A. Smits, B. de Vries, I. Weites, T. G. G. Groothuis, and H. Woelders. 2013. Yolk concentrations of hormones and glucose and egg weight and egg dimensions in unincubated chicken eggs, in relation to egg sex and hen body weight. *Gen. Comp. Endocrin.* 187:15–22.
- Aslam, M. A., T. G. G. Groothuis, M. A. Smits, and H. Woelders. 2014. Effect of corticosterone and hen body mass on primary sex ratio in laying hen (*Gallus gallus*), using unincubated eggs. *Biol. Reprod.* 90:1–9.
- Aslam, M. A., D. Schokker, T. G. G. Groothuis, A. A. C. de Wit, M. A. Smits, and H. Woelders. 2015. Association of egg mass and egg sex: gene expression analysis from maternal RNA in the germinal disc region of layer hens (*Gallus gallus*). *Biol. Reprod.* 92:1–9.

- Bartels, T., B. Fischer, P. Krüger, E. Koch, M. Ryll, and M.-E. Krautwald-Junghanns. 2008. 3D-Röntgen-Mikrocomputertomographie und Optische Kohärenztomographie als Methoden zur Lagebestimmung des Blastoderms im unbebrüteten Hühnerei. *Dtsch. Tierärztl. Wochenschr.* 115:182–188.
- Bartels, T., G. Steiner, G. Preusse, R. Galli, A. Förster, R. Preisinger, K. Cramer, and M.-E. Krautwald-Junghanns. 2014. Spektroskopische Methoden zur Geschlechtsfrühdiaagnose in der Legehennenvermehrung. *Rundsch. Fleischhyg. Lebensmittelüberw.* 66:440–442.
- Bjørnstad, S., L. P. E. Austdal, B. Roald, J. C. Glover, and R. E. Paulsen. 2015. Cracking the egg: potential of the developing chicken as a model system for nonclinical safety studies of pharmaceuticals. *J. Pharmacol. Exp. Ther.* 355:386–396.
- Burkhardt, A., S. Meister, R. Bergmann, and E. Koch. 2011. Influence of storage on the position of the germinal disc in the fertilized unincubated chicken egg. *Poult. Sci.* 90:2169–2173.
- Bruijns, M. R. N., V. Blok, E. N. Stassen, and H. G. J. Gremmen. 2015. Moral “lock-in” in responsible innovation: the ethical and social aspects of killing day-old chicks and its alternatives. *J. Agric. Environ. Ethics* 28:939–960.
- Buhl, A. C. 2013. Legal aspects of the prohibition on chick shredding in the German state of North Rhine-Westphalia. *Glob. J. Anim. Law* 2:1–8.
- Chaiken, J., J. Goodisman, B. Deng, R. J. Bussjager, and G. Shaheen. 2009. Simultaneous, noninvasive observation of elastic scattering, fluorescence and inelastic scattering as a monitor of blood flow and hematocrit in human fingertip capillary beds. *J. Biomed. Opt.* 0001;14:050505–050505–3.
- Clinton, M., S. Nandi, D. Zhao, S. Olson, P. Peterson, T. Burdon, and D. McBride. 2016. Real-time sexing of chicken embryos and compatibility with in ovo protocols. *Sex. Dev.* 10:210–216.
- Collins, K. E., B. J. Jordan, B. L. McLendon, K. J. Navara, R. B. Beckstead, and J. L. Wilson. 2013. No evidence of temperature-dependent sex determination or sex-biased embryo mortality in the chicken. *Poult. Sci.* 92:3096–3102.
- Damme, K., and M. Ristic. 2003. Fattening performance, meat yield and economic aspects of meat and layer type hybrids. *World. Poult. Sci. J.* 59:49–51.
- Davenel, A., P. A. Eliat, S. Quellec, and Y. Nys. 2015. Attempts for early gender determination of chick embryos in ovo using magnetic resonance imaging. *World. Poult. Sci. J.* 71(Suppl. 1):100.
- Doran, T. J., K. R. Morris, T. G. Wise, T. E. O’Neil, C. A. Cooper, K. A. Jenkins, and M. L. V. Tizard. 2017. Sex selection in layer chickens. *Anim. Prod. Sci.* <http://dx.doi.org/10.1071/AN16785>.
- Eiby, Y. A., J. Worthington Wilmer, and D. T. Booth. 2008. Temperature-dependent sex-biased embryo mortality in a bird. *Proc. Biol. Sci.* 275:2703–2706.
- Eide, A. L., and J. C. Glover. 1995. Development of the longitudinal projection patterns of lumbar primary sensory afferents in the chicken embryo. *J. Comp. Neurol.* 353:247–259.
- Eide, A. L., and J. C. Glover. 1997. Developmental dynamics of functionally specific primary sensory afferent projections in the chicken embryo. *Anat. Embryol. (Berl.)* 195:237–250.
- Ellendorff, F., and S. Klein. 2003. Current knowledge on sex determination and sex diagnosis potential solutions. *World. Poult. Sci. J.* 59:5–6.
- Fineman, R. M., G. C. Schoenwolf, M. Huff, P. L. Davis, and D. J. Prieur. 1986. Animal model: Causes of windowing-induced dysmorphogenesis (neural tube defects and early amnion deficit spectrum) in chicken embryos. *Am. J. Med. Genet.* 25:489–505.
- Galli, R., G. Preusse, O. Uckermann, T. Bartels, M.-E. Krautwald-Junghanns, E. Koch, and G. Steiner. 2016. In ovo sexing of domestic chicken by Raman spectroscopy. *Anal. Chem.* 88:8657–8663.
- Galli, R., G. Preusse, O. Uckermann, T. Bartels, M.-E. Krautwald-Junghanns, E. Koch, and G. Steiner. 2017a. In-ovo sexing of chicken eggs by fluorescence spectroscopy. *Anal. Bioanal. Chem.* 409:1185–1194.
- Galli, R., E. Koch, G. Preusse, C. Schnabel, T. Bartels, M.-E. Krautwald-Junghanns, and G. Steiner. 2017b. Contactless in ovo sex determination of chicken eggs. *Curr. Direct. Biomed. Eng.* 3:131–134.
- Galli, R., G. Preusse, C. Schnabel, T. Bartels, K. Cramer, M.-E. Krautwald-Junghanns, E. Koch, and G. Steiner. 2017c. Sexing of chicken eggs by fluorescence and Raman spectroscopy through the shell membrane. *PLoS One*, in press.
- Gerken, M., D. Jaenecki, and M. Kreuzer. 2003. Growth, behaviour and carcass characteristics of egg-type cockerels compared to male broilers. *World. Poult. Sci. J.* 59:45–48.
- Göhler, D., B. Fischer, and S. Meissner. 2017. In-ovo sexing of 14-day-old chicken embryos by pattern analysis in hyperspectral images (VIS/NIR spectra): A non-destructive method for layer lines with gender-specific down feather color. *Poult. Sci.* 96:1–4.
- Göth, A., and D. T. Booth. 2005. Temperature-dependent sex ratio in a bird. *Biol. Lett.* 1:31–33.
- Hardy, I. C. W. 2003. Factors influencing avian sex ratios. *World. Poult. Sci. J.* 59:18–23.
- Harz, M., M. Krause, T. Bartels, K. Cramer, P. Rösch, and J. Popp. 2008. Minimal invasive gender determination of birds by means of UV-resonance Raman spectroscopy. *Anal. Chem.* 80:1080–1086.
- Icken, W., M. Schmutz, D. Caverio, and R. Preisinger. 2013. Dual purpose chicken: the breeder’s answer to the culling of day-old male layers. *Proc. 9th European Symposium on Poultry Welfare*, Uppsala, Sweden.
- Imholt, D. 2010. Morphometrische Studien an Eiern von Hybrid- und Rassehühnern mit Versuchen zur Detektion einer Beziehung zwischen der Form von Eiern und dem Geschlecht der darin befindlichen Küken. VVB Lauferweiler Verlag, Giessen.
- Kagami, H. 2003. Sex reversal in chicken. *World. Poult. Sci. J.* 59:14–17.
- Kaleta, E. F., and T. Redmann. 2008. Approaches to determine the sex prior to and after incubation of chicken eggs and of day-old chicks. *World. Poult. Sci. J.* 64:391–399.
- Klein, S., M. Rokitta, U. Baulain, J. Thielebein, A. Haase, and F. Ellendorff. 2002. Localization of the fertilized germinal disc in the chicken egg before incubation. *Poult. Sci.* 81:529–536.
- Klein, S., U. Baulain, M. Rokitta, G. Marx, J. Thielebein, and F. Ellendorff. 2003a. Sexing the freshly laid egg – development of embryos after manipulation; analytical approach and localization of the blastoderm in the intact egg. *World. Poult. Sci. J.* 59:38–44.
- Klein, S., D. Flock, and F. Ellendorff. 2003b. Management of newly hatched male layer chicks – current knowledge on sex determination and sex diagnosis in chicken, potential solutions. *World. Poult. Sci. J.* 59:60–62.
- Koenig, M., G. Hahn, K. Damme, and M. Schmutz. 2010. Utilization of laying type cockerels as coquelets - growth performance and carcass quality. *Fleischwirtschaft* 90:92–94.
- Koenig, M., G. Hahn, K. Damme, and M. Schmutz. 2012a. Utilization of laying type cockerels as “coquelets”: influence of genotype and diet characteristics on growth performance and carcass composition. *Arch. Geflügelk.* 76:197–202.
- Koenig, M., G. Hahn, K. Damme, and M. Schmutz. 2012b. Untersuchungen zur Mastleistung und Schlachtkörperzusammensetzung von Stubenküken aus verschiedenen Legehybridherkünften. *Züchtungskunde* 6:511–522.
- Krautwald-Junghanns, M.-E., T. Bartel, K. Cramer, A. Einspanier, B. Fischer, A. Förster, R. Galli, E. Koch, S. Meissner, G. Preusse, R. Preisinger, G. Steiner, and A. Weissmann. 2014. Tötung männlicher Eintagsküken aus Legehennenlinien - Forschungsansätze für Alternativen. *DTB* 9:1228–1232.
- Krautwald-Junghanns, M.-E., T. Bartel, K. Cramer, B. Fischer, A. Förster, R. Galli, M. Huchler, S. Meissner, G. Preusse, R. Preisinger, and G. Steiner. 2015. Spektroskopische Geschlechtsbestimmung im Hühnerei. *Proc. 89th Fachgespräch über Geflügelkrankheiten*, Hannover, Verlag der DVG-Service GmbH, Gießen, pp. 17–19.
- Leiqing, P., Z. Wei, Y. Minli, S. Ye, G. Xinzhe, M. Long, L. Zijun, H. Pengcheng, and T. Kang. 2016. Gender determination of early chicken hatching eggs embryos by hyperspectral imaging. *Trans. Chin. Soc. Agric. Engin.* 32:181–186.
- Morita, V. S., I. C. Boleli, and A. Cargnelutti. 2009. Hematological values and body, heart and liver weights of male and female broiler embryos of young and old breeder eggs. *Braz. J. Poult. Sci.* 11:7–15.
- Nandi, S., D. McBride, R. Blanco, and M. Clinton. 2003. Sex diagnosis and sex determination. *World. Poult. Sci. J.* 59:7–13.
- Ort, J. -D. 2010. Zur Tötung unerwünschter neonater und juveniler Tiere. *NuR* 2010:853–861.

- Phelps, P., A. Bhutada, S. Bryan, A. Chalker, B. Ferrell, S. Neuman, C. Ricks, H. Tran, and T. Butt. 2003. Automated identification of male layer chicks prior to hatch. *World. Poult. Sci. J.* 59:32–37.
- Poultry Site. 2015. Global poultry trends 2014: rapid growth in Asia's egg output. Accessed Jul. 2016. <http://www.thepoultrysite.com/articles/3446/global-poultry-trends-2014-rapid-growth-in-asias-egg-output/>.
- Preisinger, R. 2003. Sex determination in poultry – a primary breeder's view. *World. Poult. Sci. J.* 59:52–56.
- Preisinger, R., W. Icken, and M. Schmutz. 2014. Breeding dual-purpose chicken opposed to specialised hybrids. *Proc. XIVth EuropeanPoultry Conference, Stavanger (Norwegen), 23.06.-27.06.2014*, S140.
- Quansah, E. S., N. A. R. Urwin, P. Strappe, and S. Raidal. 2013. Progress towards generation of transgenic lines of chicken with a green fluorescent protein gene in the female specific (w) chromosome by sperm-mediated gene transfer. *Adv. Genet. Eng.* 2:29.
- Rosenbruch, M. 1994. Frühe Entwicklungsstadien des bebrüteten Hühnereies als Modell in der experimentellen Biologie und Medizin. *ALTEX* 11:199–206.
- Rosenbruch, M. 1997. Zur Sensitivität des Embryos im bebrüteten Hühnerei. *ALTEX* 14:111–113.
- Rozenboim, I., and E. Ben Dor. 2011. The use of reflectance spectroscopy for fertility detection in freshly laid egg and gender sorting in mid incubation period. *Poult. Sci.* 90(E-suppl. 1):98. (Abstr.).
- Seemann, G. 2003. Organisational framework for hatcheries. *World. Poult. Sci. J.* 59:57–59.
- Smith, C.- A., K. N. Roeszler, Q. J. Hudson, and A. H. Sinclair. 2007. Avian sex determination: what, when and where? *Cytogenet. Genome Res.* 117:165–173.
- Steiner, G., T. Bartels, A. Stelling, M.-E. Krautwald-Junghanns, H. Fuhrmann, V. Sablinskas, and E. Koch. 2011. Gender determination of fertilized unincubated chicken eggs by infrared spectroscopic imaging. *Anal. Bioanal. Chem.* 400:2775–2782.
- Tiersch, T. R. 2003. Identification of sex in chickens by flow cytometry. *World. Poult. Sci. J.* 59:24–31.
- Tran, H. T., W. Ferrell, and T. R. Butt. 2010. An estrogen sensor for poultry sex sorting. *J. Anim. Sci.* 88:1358–1364.
- Tyack, S. G., K. A. Jenkins, T. E. O'Neil, T. G. Wise, K. R. Morris, M. P. Bruce, S. McLeod, A. J. Wade, J. McKay, R. J. Moore, K. A. Schat, J. W. Lowenthal, and T. J. Doran. 2013. A new method for producing transgenic birds via direct in vivo transfection of primordial germ cells. *Transgenic Res.* 22:1257–1264.
- Webster, B., W. Hayes, and T. W. Pike. 2015. Avian egg odour encodes information on embryo sex, fertility and development. *PLoS One* 10:e0116345. doi:10.1371/journal.pone.0116345.
- Weissmann, A., S. Reitemeier, A. Hahn, J. Gottschalk, and A. Einspanier. 2013. Sexing domestic chicken before hatch: a new method for in ovo gender identification. *Theriogenology* 80:199–205.
- Weissmann, A., A. Förster, J. Gottschalk, S. Reitemeier, M. - E. Krautwald-Junghanns, R. Preisinger, and A. Einspanier. 2014. In ovo-gender identification in laying hen hybrids: effects on hatching and production performance. *Eur. Poult. Sci.* 78:199–205.
- Whittaker, D. J., H. A. Soini, N. M. Gerlach, A. L. Posto, M. V. Novotny, and E. D. Ketterson. 2011. Role of testosterone in stimulating seasonal changes in a potential avian chemosignal. *J. Chem. Ecol.* 37:1349–1357.
- Yilmaz-Dikmen, B., and S. Dikmen. 2013. A morphometric method of sexing white layer eggs. *Braz. J. Poult. Sci.* 15:203–210.